

# Proposed Project Charter: MEP AI Scheduler

## Problem Statement

### Strategic Context

Lazenby Electric aims to evolve from a single-trade electrical contractor into a full-service MEP (Mechanical, Electrical, Plumbing) provider. The strategic goal is to offer General Contractors a single point of contact for all major systems in residential and commercial builds. However, current scheduling operations cannot support this scale. Scheduling is performed manually on a physical whiteboard by senior partners. This process involves high-level mental gymnastics to balance job hierarchy, employee skill sets, logistical constraints, and daily volatility. The current manual, whiteboard-based scheduling system drains executive time, lacks real-time agility, and presents a major scalability barrier that prevents the targeted expansion.

## Proposed AI Solution Design

A Constraint-Based adaptive scheduling engine integrated with Lazenby's existing data.

### Technical Architecture

- Input Layer:
  - ProCore Integration: Automated fetching of active jobs, estimated man-hours, and budgetary constraints via the ProCore API.
  - Resource Database: A structured roster tagging every employee with skills (e.g., Journeyman Electrician, HVAC Tech II), home location, and availability.
- Core Engine:
  - Constraint Satisfaction Model (CSP): Build a solver (using Python/OR-Tools) that mathematically guarantees *hard constraints* (e.g., Job A needs 3 electricians on Tuesday) are met while optimizing *soft constraints* (e.g., minimize travel time).
- User Interface:
  - Admin Dashboard: A web interface that replaces the whiteboard, allowing partners to approve AI-generated schedules and manually override constraints.

### Data Flow

1. **Ingest:** Engine pulls Job Data (ProCore) and Employee Data (new SQL database).
2. **Process:** The runs a Solver to find the optimal schedule.
3. **Output:** The user interface suggests a weekly schedule to the Partner for approval.

## Roadmap

### Sprint 1: Data Foundation & Core Logic

- **Goal:** Be able to input data and output a valid schedule, even if not optimal.
- **Key Activities:**
  - **Data Pipeline:** Build Python script to authenticate with ProCore API and fetch list of active projects / jobsites.
  - **Digitize Rules:** Interview partners to list top 5+ hard constraints (e.g., Apprentice must be with Journeyman).

- **Prototype Solver:** Implement a basic constraint solver that assigns people to jobs without conflicts.
- **Deliverable:** A script that takes a CSV of dummy jobs/employees and outputs a conflict-free schedule.

## Sprint 2: Optimization & User Interface

- **Goal:** Apply soft constraints and wrap the engine in a usable UI.
- **Key Activities:**
  - **Optimization Layer:** Update the solver to penalize travel distance (using simple lat/long math) to optimize routing.
  - **UI Build:** Create a simple Streamlit or React frontend to display the schedule visually (Gantt chart view).
  - **Validation:** Run the AI schedule against last week's whiteboard schedule to compare efficiency.
- **Deliverable:** Functional minimum viable product where a user can click "Generate Optimal Schedule," and receive a suggestion for the best schedule for next week.

## Estimated ROI

### Cost of Status Quo

- Executive Time: 4 partners × 5 hours/week × \$200/hr = \$4,000/week → \$208,000/year
- Efficiency Leaks: Misallocation of labor (e.g., excessive travel or wrong crew mix) estimated at 2% of total labor budget. Assuming, 20 employees × \$40/hr × 40 hrs/week = \$32,000 weekly payroll, then 2% leakage = \$640/week → \$33,280/year
- **Total Annual Cost of Problem:** (\$4,640/week) × 52 weeks = **\$241,280/year.**

### Estimated Cost of Project

- **Development Labor:** 40 hours × \$100/hr market rate = **\$4,000.**
- **Hosting/Tools:** AWS/Azure credits + ProCore API access = **\$500.**
- **Total Investment:** **\$4,500.**

### Financial Analysis

- **Payback period:** <1 week
- Tangible Savings
  - 80% reduction in partner time: \$166,400/year
  - 50% reduction in efficiency leaks 50%: \$16,600/year
  - **Total Annual Savings: \$183,000/year**
- Year-1 Net Benefit: \$183,000 - \$4,500 = \$178,500
- **5-year NPV: \$888,000**